DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UNIVERSITY OF BARISHAL
FINAL EXAMINATION

Course Title: Mathematical Analysis for Computer Science
Course Code: CSE-3201
3rd Year 2nd Semester
Session: 2016-17(Admission)
(Answer Any Five Questions)

Time: 3 Hours Marks: 60

1. a) Write down the reason of studying mathematical analysis in Computer Science and Engineering. [2]

   b) Define standard deviation and variance. In a certain set of numbers, 12.5 is 1.5 units of standard deviation above the mean, and 8.9 is 0.5 units of standard deviation below the mean. What is the mean of the set? [2.5]

   c) What do you mean by conditional probability? A box contains 4 red chips and 2 blue chips. If two chips are selected at random without replacement, what is the probability that the chips are different colors? [3.5]

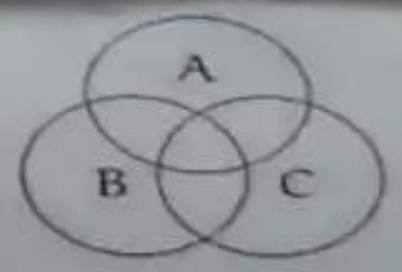
   d) Probability is the measure of the likelihood that an event will occur.Probability quantifies as a number between 0 and 1, where, loosely speaking, 0 indicates impossibility and 1 indicates certainty. [4]
      i. A die is rolled, find the probability that the number obtained is greater than 4.
      ii. Two coins are tossed, find the probability that one head only is obtained.
      iii. Two dice are rolled, find the probability that the sum is equal to 5.
      iv. A card is drawn at random from a deck of cards. Find the probability of getting the King of heart.

2. a) For all positive integers x, the function f is defined as follows: [2]

$$f(x) = \begin{cases} x & if\ x\ is\ a\ prime \\ x-1 & if\ x\ is\ not\ prime \end{cases}$$

   If, $f(a)\ x\ f(a+1) = a^2$; write down the conditions for which the overall conditions hold.

   b) A "Venn diagram" with three overlapping circles is often used to illustrate the eight possible subsets associated with three given sets, Can the sixteen possibilities that arise with four given sets be illustrated by four overlapping circles? Justify your answer. [3]



   c) Write down the recursive formula and complexity analysis of the Tower of Hanoi Problem. [3]

   d) Using taylor's series method, find y at x=0.1,0.2 correct up to three significant digits by solving $\frac{dy}{dx} = x - y\ with\ y(0) = 1$. [4]

3. a) Set Q contains 20 positive integer values. The smallest value in Set Q is a single digit value and the largest value in Set Q is a three digit value. What are possible values for the range of Set Q? Find out the minimum and maximum range for set Q.[Range in a set means, $Range_Q = Q_{max} - Q_{min}$.] [2]

   b) Write down the recursive formula for the following problems. [4]
      i. Fibonacci Number
      ii. Factorial Number
      iii. GCD

   c) A number in decimal notation is divisible by 3 if and only if the sum of its digits is divisible by 3. [3]
   Prove this well-known rule, and generalize it.

   d) The length of human pregnancies from conception to birth approximates a normal distribution with a mean of 266 days and a standard deviation of 16 days. What proportion of all pregnancies will last between 240 and 270 days (roughly between 8 and 9 months)? [3]

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